



Solar Powered Distress Beacon

The Solderers:
Joshua Bisdorf, Jonathan Duke, Garrett Merbach, and Ryan Parks
Presentation on Saturday, April 15, 2023

Background Information

Our theme is remote sensor

The outdoors can be very dangerous

No solar powered options widely available

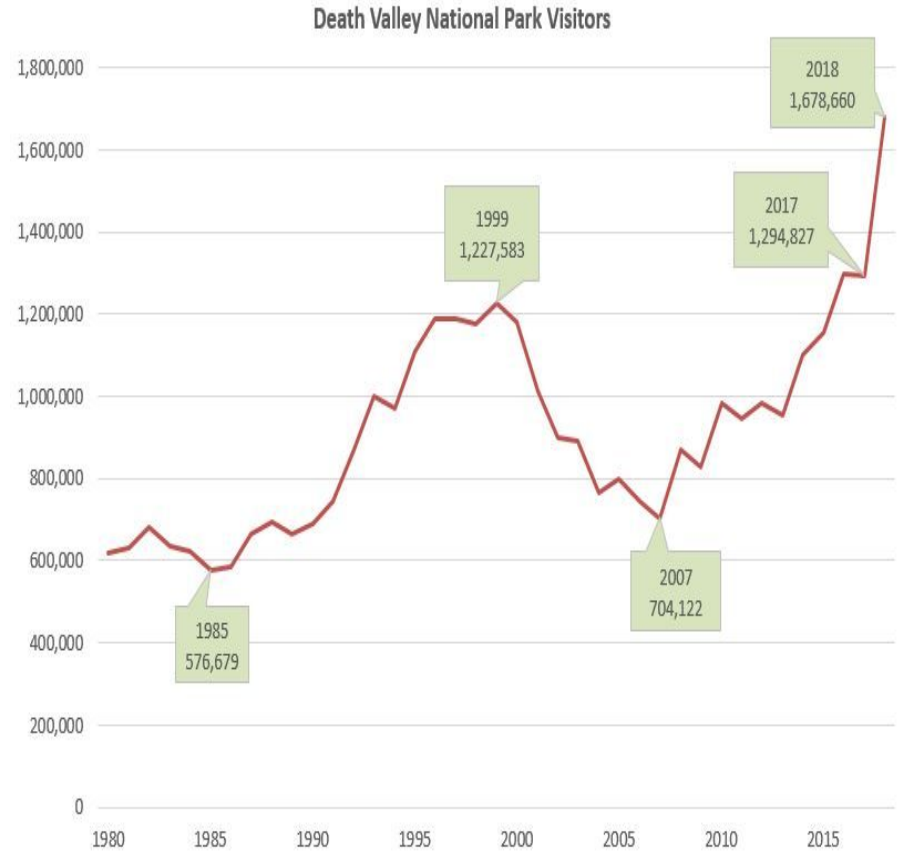


Figure 2: “Death Valley National Park Visitors” [2]

Images of Prototype

Dimensions: (19.5cm x 12.5cm x 5cm)

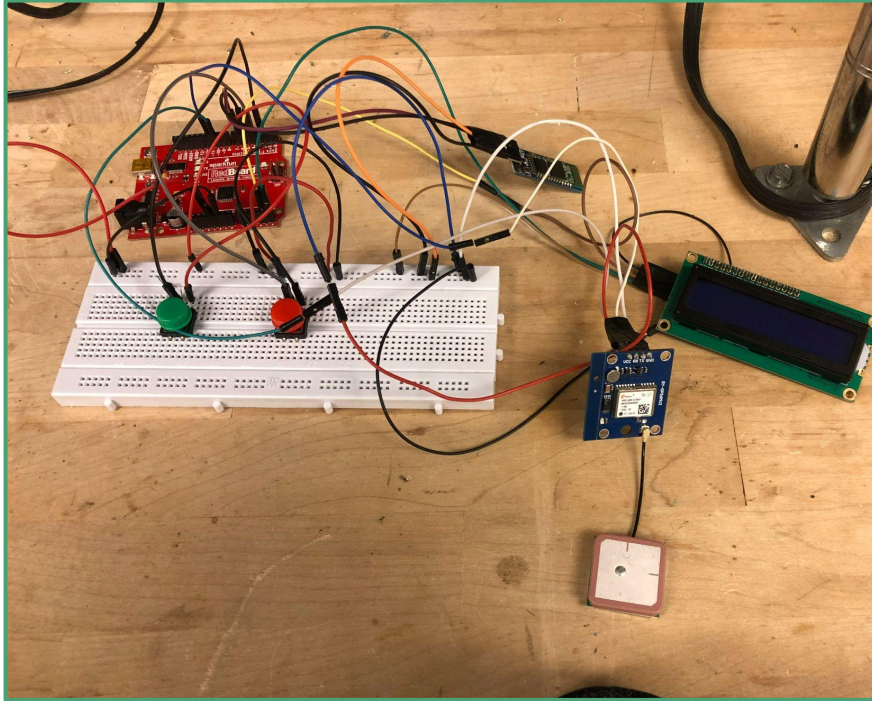


Figure 3: The Circuit of the Distress Beacon

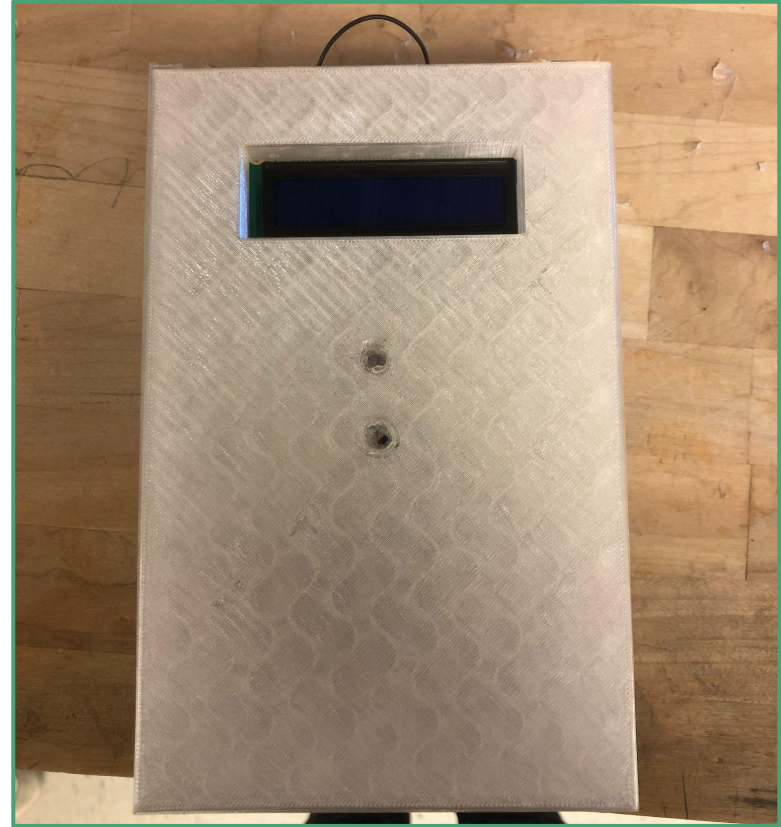


Figure 4: Casing of the Distress Beacon

Overview of Prototype

Charge with no assistance

Request data to the user

Send data to a receiving station

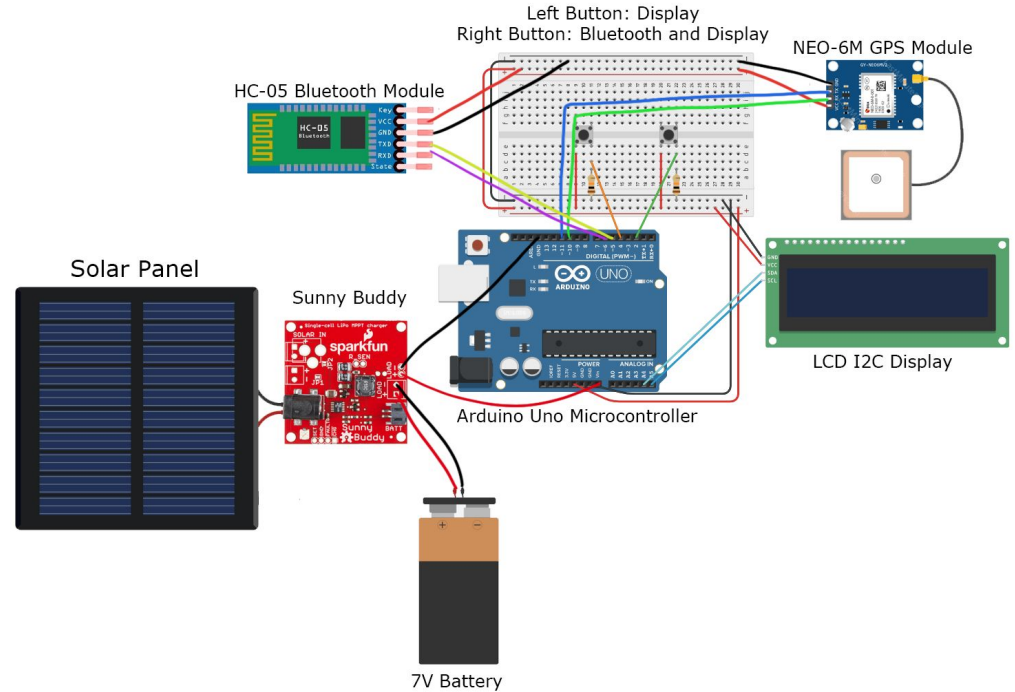


Figure 5: Schematic Created at [tinkercad.com](https://www.tinkercad.com)

Technical Requirements

Prototype

Pair bluetooth module with mobile device and transmit data, up to 210 ft

Requires 0.27W to run while all components are turned on

Solar panel provides power for components with 0.156W surplus

16 x 2 LCD Display precise to 2 decimal places

Real World

Send SOS signal with frequency of 406 MHz to SARTSAT satellite system with worldwide coverage [3]

Needs to emit 5W signal and 0.17W to run other components [4]

Solar panel able to provide enough power for other components but not satellite signal

Bigger display to allow for more precision (ex 128x64 OLED)

Testing Procedures and Results

	GPS	Bluetooth	Display	Solar Panel
Test	Accuracy of coordinate location	Distance of bluetooth connection	Display data on a digital screen and update it	Power beacon with excess power to store in a battery
Results	Accurately records coordinate data up to 4 decimal places	Connects and sends data 200+ feet	Displays data for seven seconds at a time per button press	Produces little wattage compared to what is consumed

Summary of Power Usage

	Wattage (W)
GPS	0.225 [5]
Bluetooth	0.04 [6]
Display	0.005 [7]
Total Consumed	0.27
Solar Panel	0.426 (produced)

Battery Charge Time

Charge time of 22.5 hours
(while device is not running)

Lasts 35.6 hours
(without any power from solar panel)

Summary of Prototype Design Tradeoffs

First Prototype

(19.5cm X 12.5cm) Solar Panel

HC-05 Bluetooth Transmission

PLA Plastic Housing

9.6 Wh battery

Next Prototype

(11.5cm x 11.5cm) Solar Panel [8]

COSPAS-SARSAT 406MHz module [9]

Weatherproof Acrylic Plastic [10]

120 Wh battery [11]

Roadblocks to Commercialization

95,000 units sold [12]
57.8 million hikers [13]

Simpler alternatives

Not feasible with modern technology

Reliance on SARTSAT



Figure 6: Competitor's Personal Locator Beacon [14]

Value Proposition

Utility to our Stakeholders

Safety and Reliability

Uniqueness

Multi-Use



Drawbacks of our product

Charge Time

Weight

Lack of Marketability

Investment in our product is not currently recommended

Conclusions

Safety is a necessity in the wild

A life changing advancement

Send GPS location data to receiver

Drawbacks prevent recommendation

bisdorfj@umich.edu

jonpduke@umich.edu

rhparcs@umich.edu

gmerbach@umich.edu



Figure 7: “Camping Under Stars” [15]

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